UNIVERSITY OF PITTSBURGH INSITES INFRASTRUCTURE SENSING for INTELLIGENT TRANSPORTATION and ENERGY SYSTEMS CONSORTIUM

University of Pittsburgh

Research Association Position Available at University of Pittsburgh: "Applied Electromagnetics for Non-Destructive Evaluation Applications"

Position Opening: Available Immediately
Position: Applied Electromagnetics for Non-Destructive Evaluation Applications (PhD or MS)
Position Contact: Prof. Paul Ohodnicki, pro8@pitt.edu
Anticipated Position Start Date: Fall 2026

Position Description:
The INfrastructure Sensing and Intelligent Transportation Systems (INSITES) Consortium at the University of Pittsburgh invites applications for a research associate with an emphasis on applied electromagnetic modeling and experimentation for the application of non-destructive evaluation methods and techniques. Near-term anticipated project work will include modeling of electromagnetic phenomena such as guided waves, antennas, and interactions with surrounding media as well as experimental investigations of the same. The primary application focus will be on structural health monitoring and non-destructive evaluation methods, including quantitative benchmarking as compared to alternative methods and techniques.

The position is funded through the newly established INSITES consortium at the University of Pittsburgh and is a focal area of growth within the initiative creating ample opportunity for high impact scientific and engineering research with career growth potential for the successful candidate. The research associate will have an opportunity to collaborate with an interdisciplinary group of undergraduate, graduate, and PhD level researchers in advanced sensing concepts, advanced analytics using methods such as machine learning and artificial intelligence, and system level modeling including digital twins and related technologies. In addition, the research associate can expect close collaboration with other researchers from industry and national labs including in collaboration with partners at Lawrence Livermore National Laboratory, National Energy Technology Laboratory, and others.

The advisor will be Prof. Paul Ohodnicki and the research associate will have full access to facilities available in the Ohodnicki Lab (https://www.engineering.pitt.edu/OhodnickiLab/) and extensive shared facilities of the INfrastructure Sensing Technology for Intelligent Transportation Systems (INSITES, https://www.engineering.pitt.edu/INSITES) consortium, amongst other facilities across campus.

More information about the Ohodnicki Lab research focus and interests can be found here:
https://www.youtube.com/watch?v=7Vn7XJmHGr4&feature=youtu.be

Successful applicants should display strong experience in concepts of importance for applied electromagnetics modeling and design as well as experimental practice related to the same, including concepts and tools such as finite element modeling, antenna design, microwave measurements and instrumentation, and related signal processing techniques. Applicants should have a graduate degree (MS with related experience or a PhD) in electrical engineering, applied physics, or a related field.

Anticipated project assignments will include modeling of applied electromagnetic waves and wave propagation as well as antenna design and construction and related microwave measurement methods using instrumentation such as oscilloscopes, vector network analyzers, RF components, and related. In addition to conducting research, duties will also include preparing reports, performing literature reviews, supervising and mentoring students, and assisting with other projects in the laboratory. The research associate will also be responsible for establishing lab capabilities and serving as a local expert in microwave and wave propagation technology related to efforts that will occur at the Energy Innovation Center.

Application Process: Interested candidates should contact Prof. Paul Ohodnicki (pro8@pitt.edu) with an updated resume at the soonest possible opportunity.